

MATPLOTLIB

PYTHON DATA VISUALIZATION ASSIGNMENT

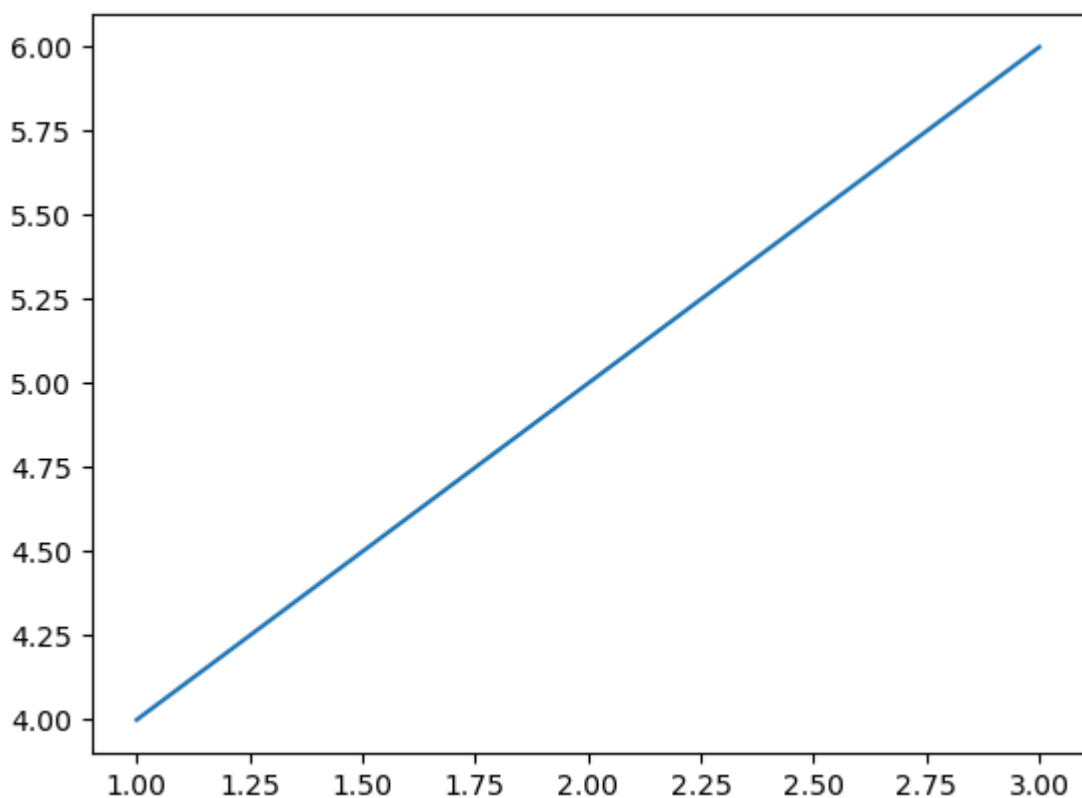
Name: Sumit Mund **SIC:** 24BCSH05 **Roll No.:** 15

1. Importing Libraries

```
In [2]: import numpy as np  
import pandas as pd  
import matplotlib.pyplot as plt
```

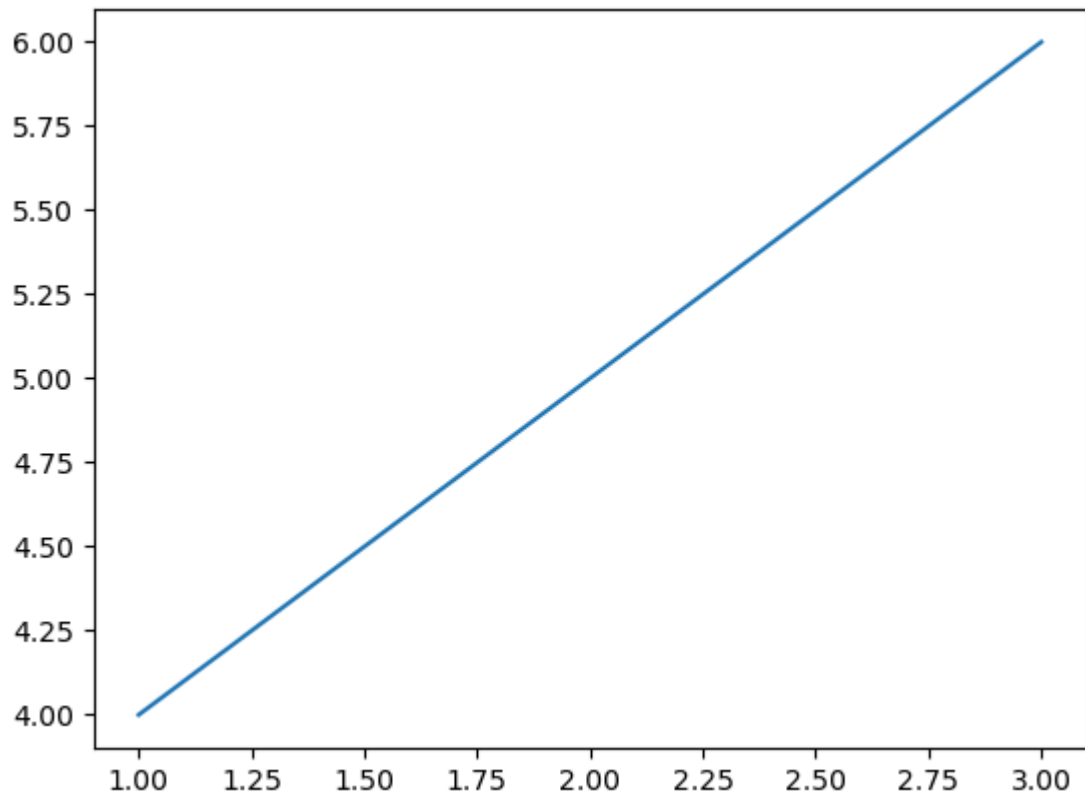
2. Simple Line Graph

```
In [3]: plt.plot([1, 2, 3], [4, 5, 6])  
plt.show()
```



3. Line Graph Using Variables

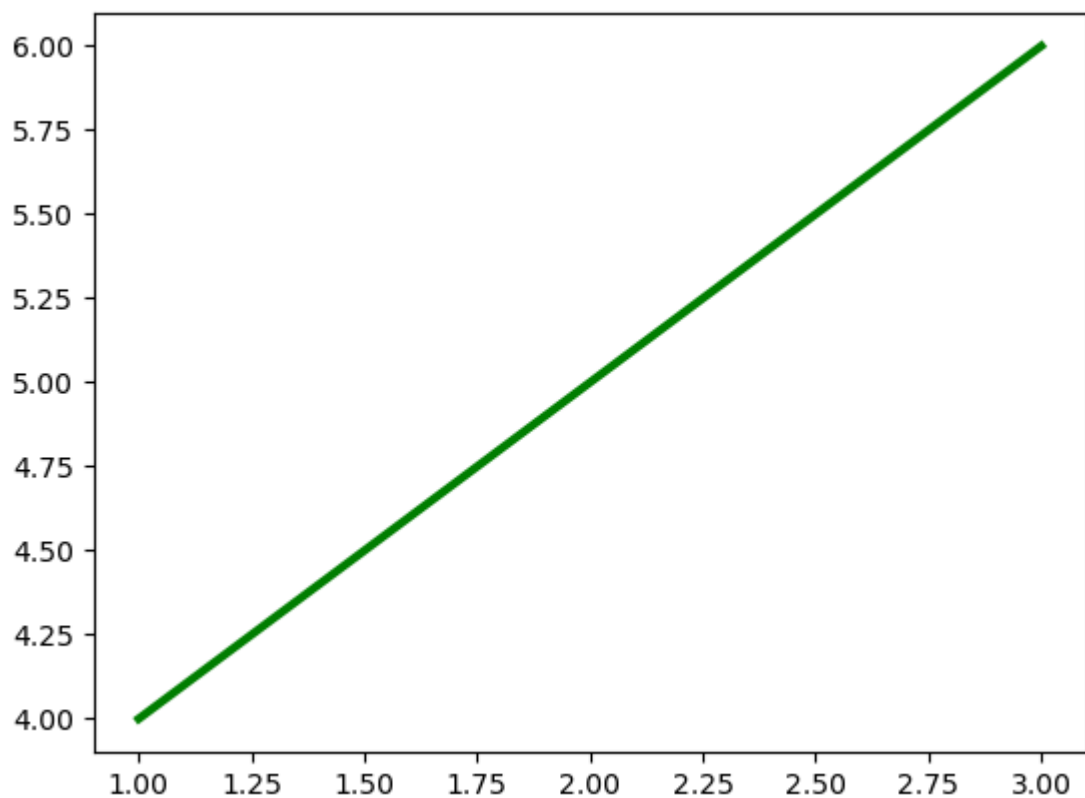
```
In [4]: x = [1, 2, 3]  
y = [4, 5, 6]  
plt.plot(x, y)  
plt.show()
```



4. Adding Color and Line Width

```
In [5]: x = [1, 2, 3]
        y = [4, 5, 6]

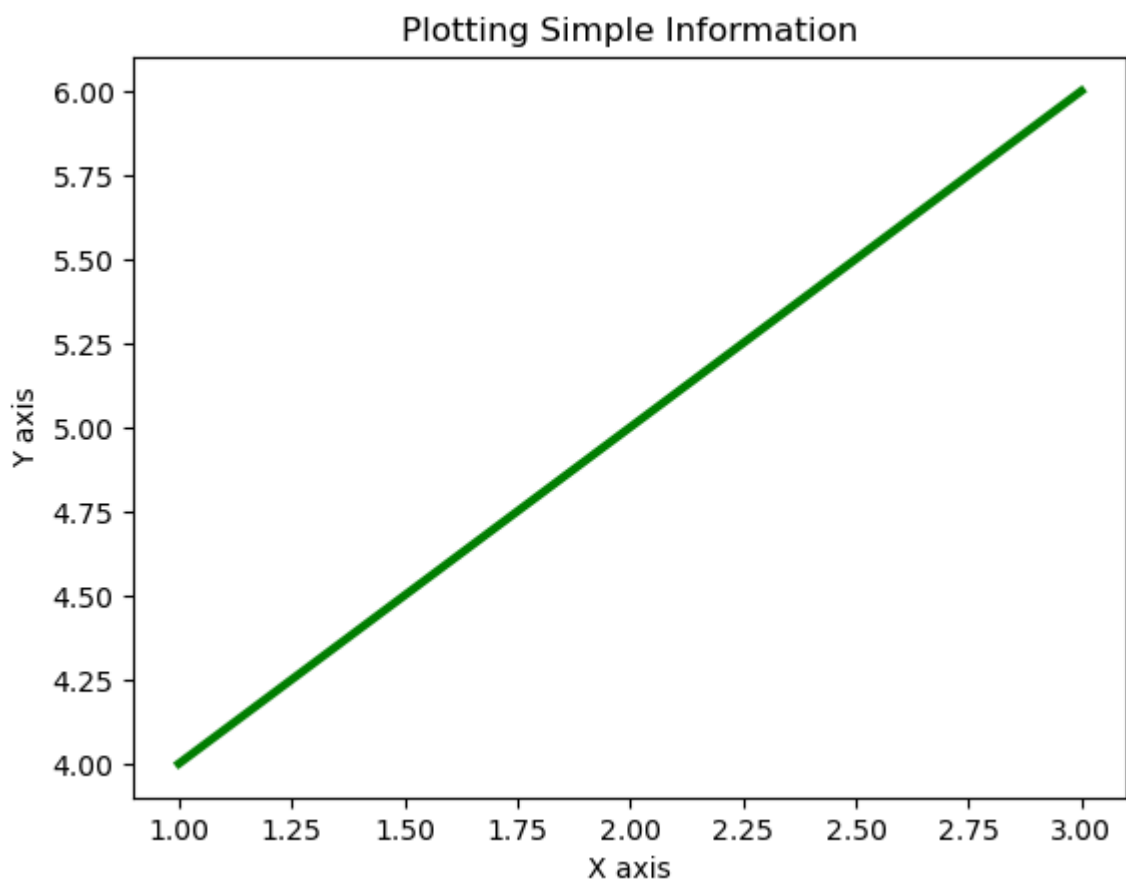
        plt.plot(x, y, color='green', linewidth=3)
        plt.show()
```



5. Title and Axis Labels

```
In [6]: x = [1, 2, 3]
y = [4, 5, 6]

plt.plot(x, y, color='green', linewidth=3)
plt.title('Plotting Simple Information')
plt.xlabel('X axis')
plt.ylabel('Y axis')
plt.show()
```



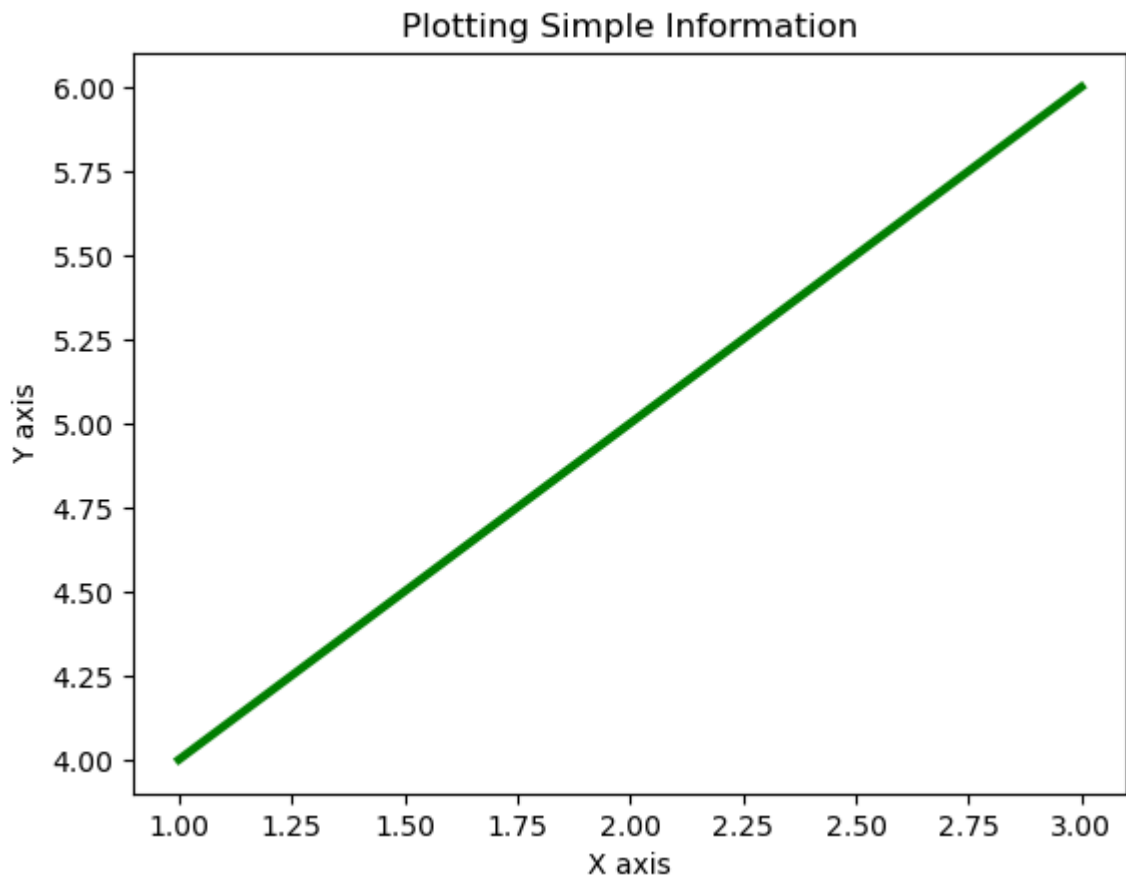
6. Saving a Graph

```
In [7]: x = [1, 2, 3]
y = [4, 5, 6]

plt.plot(x, y, color='green', linewidth=3)
plt.title('Plotting Simple Information')
plt.xlabel('X axis')
plt.ylabel('Y axis')

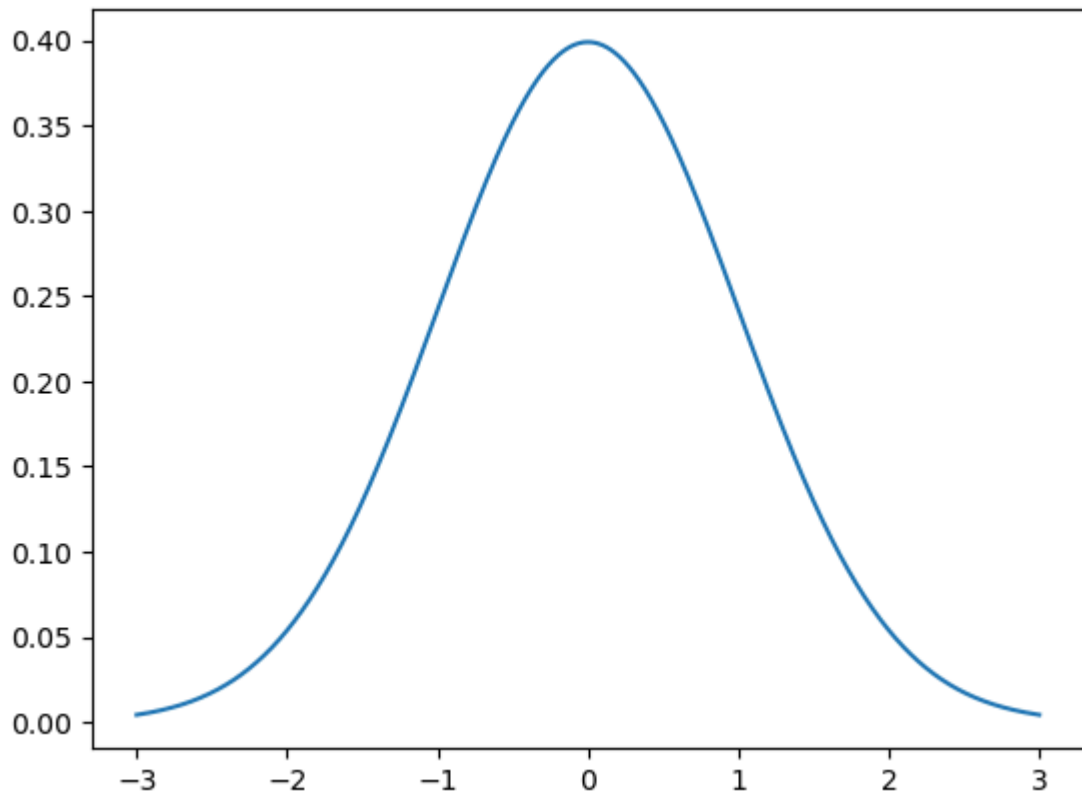
# Change this path if needed
# plt.savefig(r'C:\Users\SILICON\Desktop\24BCSH05\test.png', format='png')

plt.show()
```



7. Normal Distribution / PDF

```
In [8]: from scipy.stats import norm  
  
x = np.arange(-3, 3, 0.0001)  
  
plt.plot(x, norm.pdf(x))  
plt.show()
```



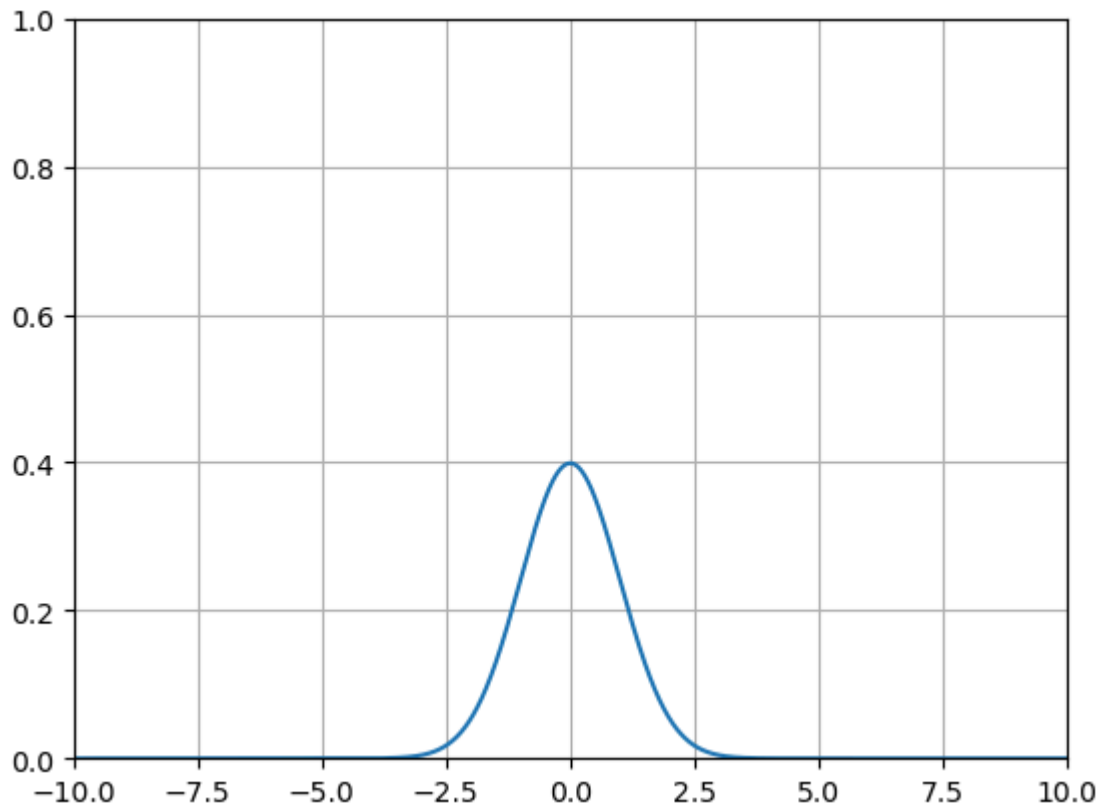
8. Setting X-axis, Y-axis Limits and Grid

```
In [9]: from scipy.stats import norm

x = np.arange(-10, 10, 0.0001)

axes = plt.axes()
axes.set_xlim([-10, 10])
axes.set_ylim([0, 1.0])
axes.grid()

plt.plot(x, norm.pdf(x))
plt.show()
```



9. Plotting Two Normal Distributions

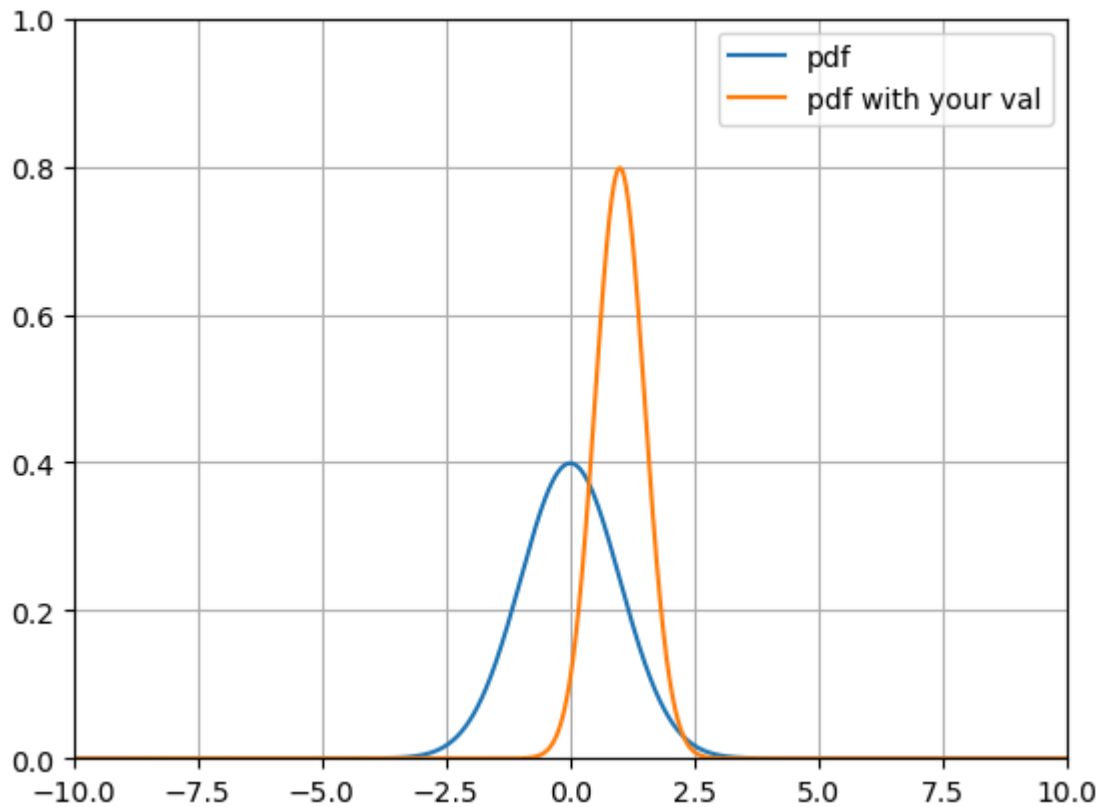
```
In [10]: from scipy.stats import norm

x = np.arange(-10, 10, 0.0001)

axes = plt.axes()
axes.set_xlim([-10, 10])
axes.set_ylim([0, 1.0])
axes.grid()

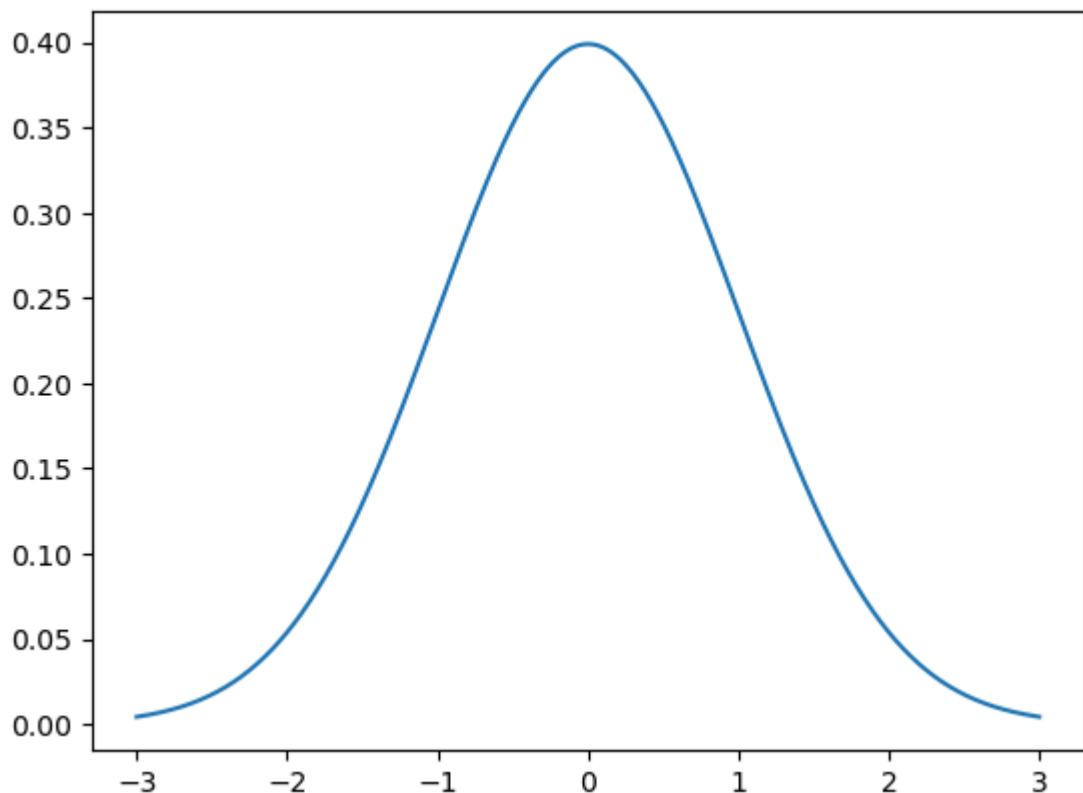
plt.plot(x, norm.pdf(x))
plt.plot(x, norm.pdf(x, 1.0, 0.5))

plt.legend(['pdf', 'pdf with your val'])
plt.show()
```



10. Draw a Line Graph Using NumPy

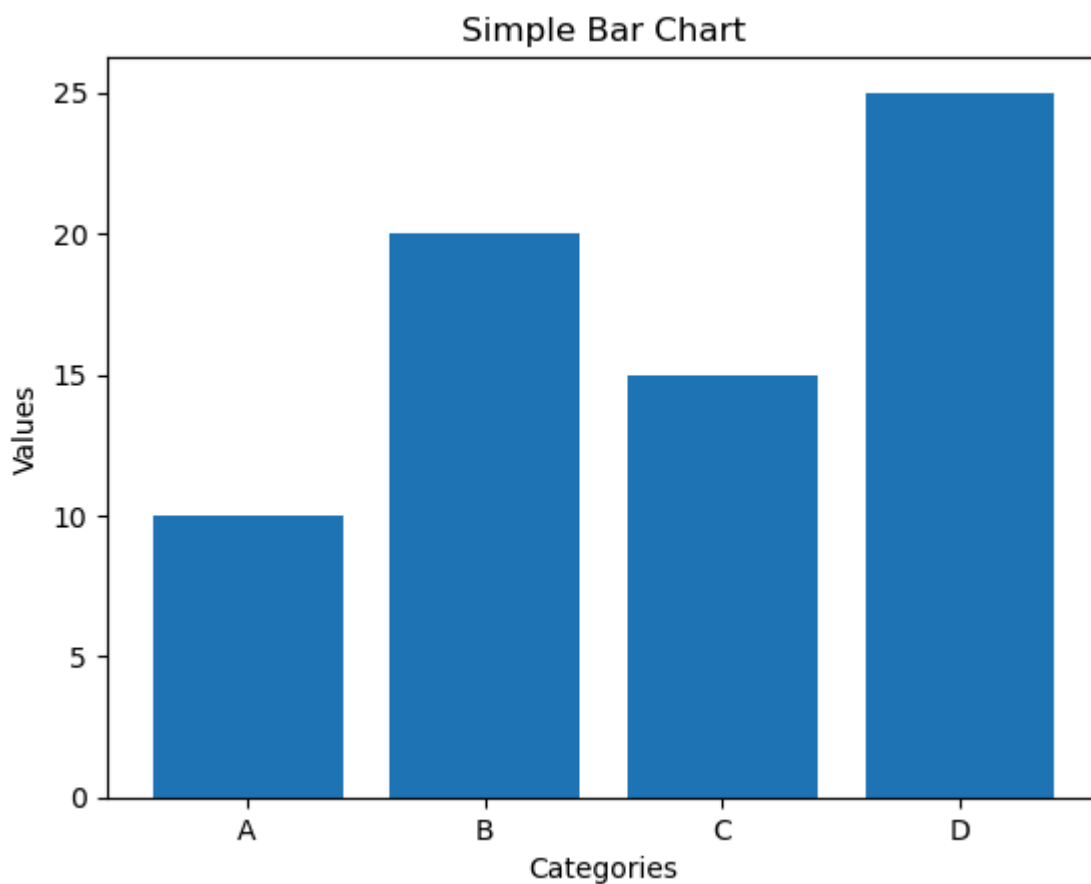
```
In [11]: x = np.arange(-3, 3, 0.0001)
plt.plot(x, norm.pdf(x))
plt.show()
```



11. Bar Chart

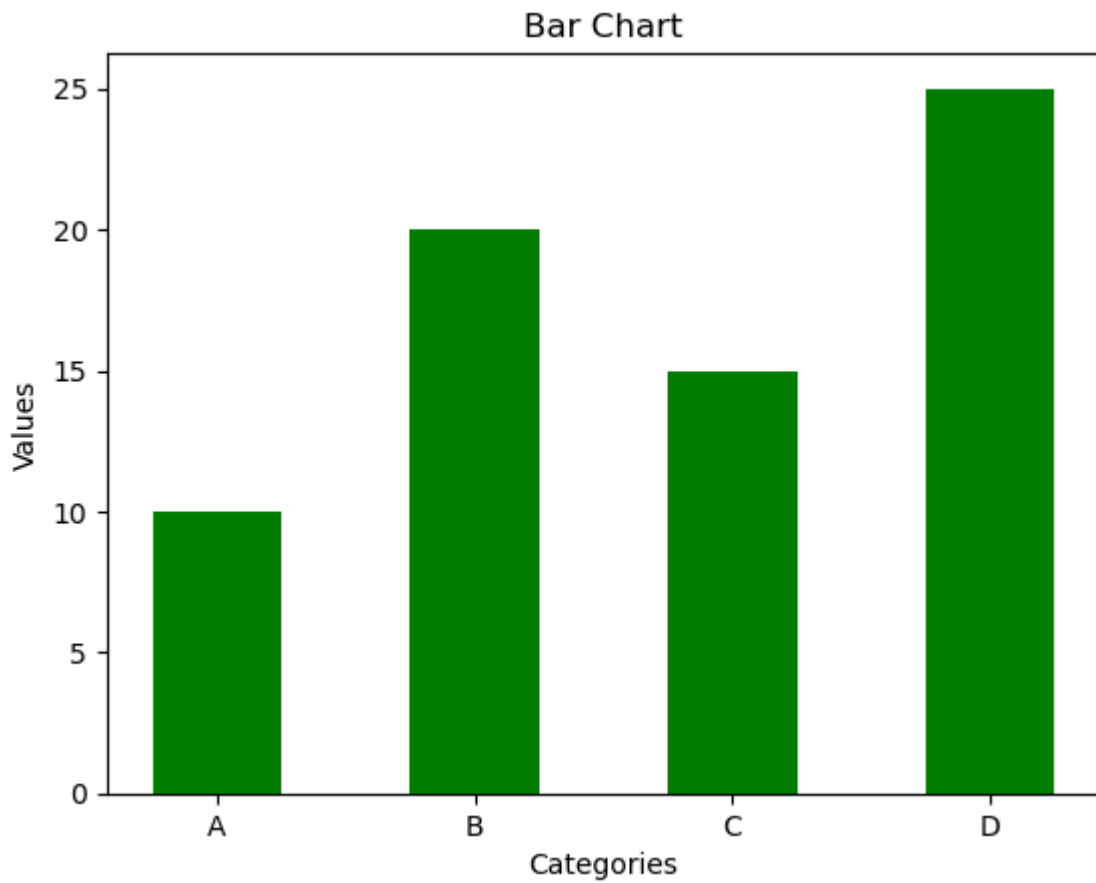
A bar chart represents categorical data using rectangular bars.

```
In [12]: x = ['A', 'B', 'C', 'D']  
y = [10, 20, 15, 25]  
  
plt.bar(x, y)  
  
plt.title('Simple Bar Chart')  
plt.xlabel('Categories')  
plt.ylabel('Values')  
  
plt.show()
```



12. Bar Chart with Color and Width

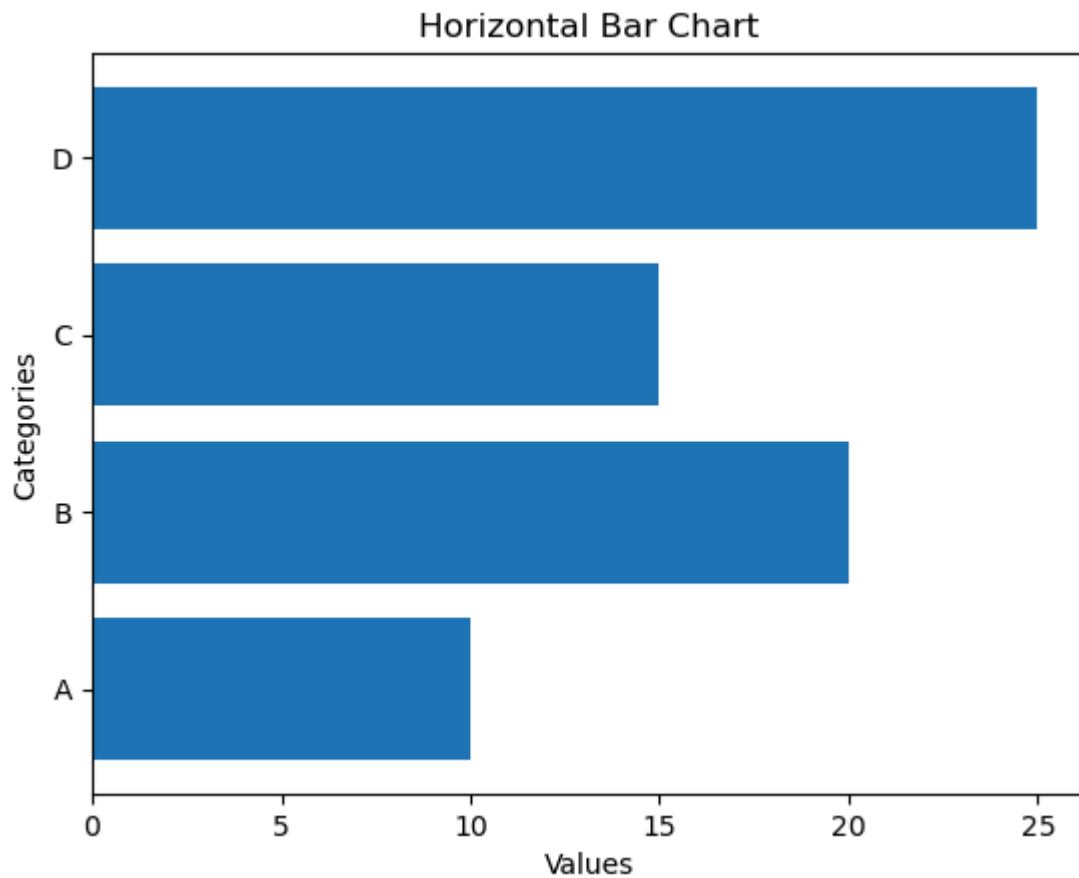
```
In [13]: x = ['A', 'B', 'C', 'D']  
y = [10, 20, 15, 25]  
  
plt.bar(x, y, color='green', width=0.5)  
  
plt.title('Bar Chart')  
plt.xlabel('Categories')  
plt.ylabel('Values')  
  
plt.show()
```



13. Horizontal Bar Chart

`plt.barh()` to create a horizontal bar chart.

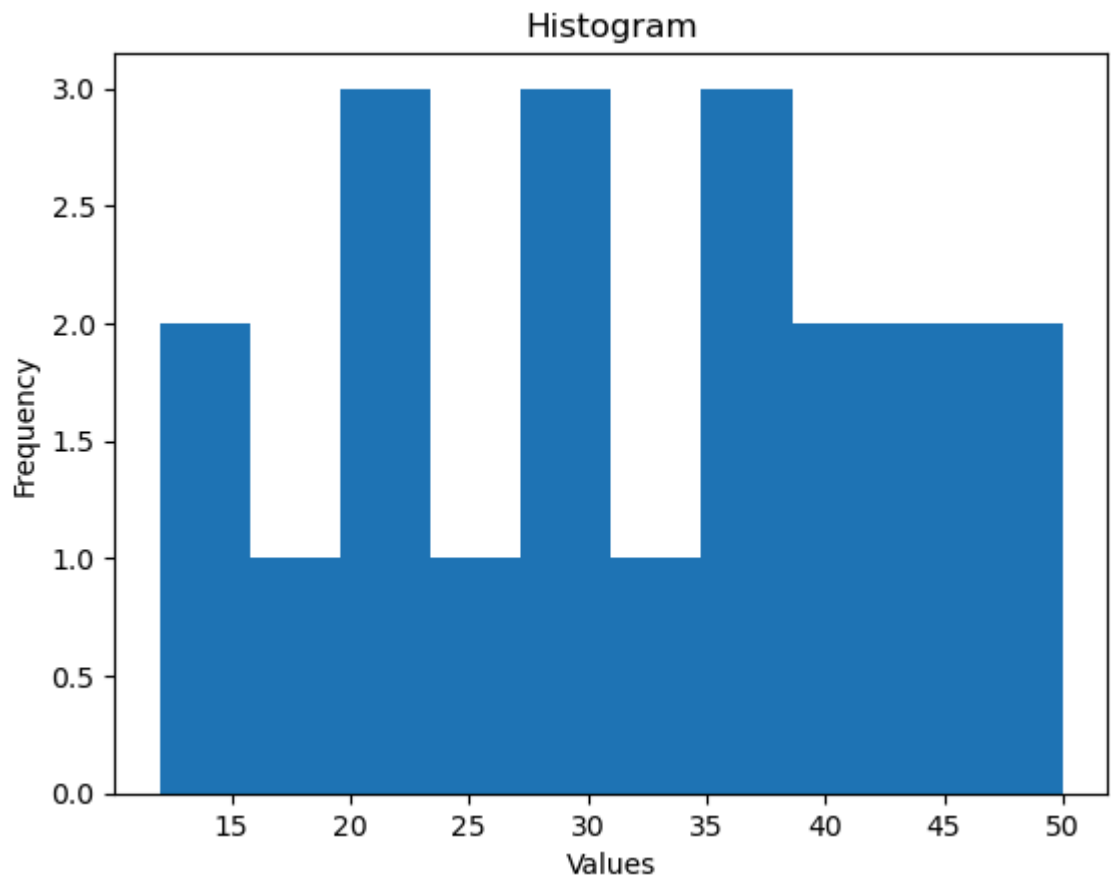
```
In [14]: x = ['A', 'B', 'C', 'D']  
y = [10, 20, 15, 25]  
  
plt.barh(x, y)  
  
plt.title('Horizontal Bar Chart')  
plt.xlabel('Values')  
plt.ylabel('Categories')  
  
plt.show()
```



14. Histogram

A histogram shows the distribution or frequency of numerical data

```
In [15]: data = [  
    12, 15, 18, 20, 22,  
    22, 25, 28, 30, 30,  
    32, 35, 35, 38, 40,  
    42, 45, 45, 48, 50  
]  
  
plt.hist(data)  
  
plt.title('Histogram')  
plt.xlabel('Values')  
plt.ylabel('Frequency')  
  
plt.show()
```



In []:

In []: